CHEMISTRY

PAPER 1 (THEORY)

Maximum Marks: 70 **Time Allowed: Three Hours**

Reading Time: Additional Fifteen Minutes

Instructions to Candidates

1. You are allowed an additional fifteen minutes for only reading the question paper.
2. You must NOT start writing during the reading time.
3. This question paper has 8 printed pages.
4. It is divided into four sections and has twenty one questions in all.
5. Answer all questions.
6. Section A has fourteen subparts. Each question carries 1 mark.
7. While attempting Multiple Choice Questions in Section A, you are required to write only ONE option as the answer.
8. Section B has ten questions. Each question carries 2 marks.
9. Section C has seven questions. Each question carries 3 marks.
10. Section D has three questions. Each question carries 5 marks.
11. Internal choices have been provided in two questions each in Sections B, C and D.
12. The intended marks for questions are given in brackets [ ].
13. All working, including rough work, should be done on the same sheet as, and adjacent to the rest of the answer.
14. Balanced equations must be given wherever possible and diagrams where they are helpful.
15. When solving numerical problems, all essential workings must be shown.
16. In working out problems, use the following data:

    Gas constant R = 1·987 cal deg^{-1} mol^{-1} = 8·314 J K^{-1} mol^{-1} = 0·0821 dm^3 atm K^{-1} mol^{-1}, 1 l atm = 1 dm^3 atm = 101·3 J.

    1 faraday = 96500 coulombs, Avogadro's number = 6·023 × 10^{23}.

Instruction to Supervising Examiner

Kindly read aloud the Instructions given above to all the candidates present in the examination hall.

SECTION A – 14 MARKS

Question 1

(A) Fill in the blanks by choosing the appropriate word(s) from those given in the brackets.

[Wolff–Kishner, Clemmensen, propane, propan-2-ol, primary, secondary, tertiary, completely filled, partly filled, diamagnetic, paramagnetic, 2,4,6-tribromophenol, 4-bromophenol, electrophilic, nucleophilic, Rosenmund]

(i) When propanone is heated with hydrazine and then with KOH in a high boiling solvent such as glycol, _____ is formed. The reaction is known as _____ reduction. *(Recall)* [1]

(ii) A dry cell is an example of a _____ cell, whereas a lead storage battery is an example of a _____ cell. *(Recall)* [1]

(iii) Transition metal ions having _____ d-orbitals are colourless and _____ in nature. [1]
(Understand)

(iv) Phenol reacts with bromine water to give _____, which is an example of _____ substitution. *(Recall)* [1]

(B) Select and write the correct alternative from the choices given below.

(i) Equimolal (0·1 m) aqueous solutions of glucose, NaCl, $CaCl_2$ and $K_3[Fe(CN)_6]$ are prepared. Assuming complete dissociation of the electrolytes, the correct order of their freezing points is: *(Apply)* [1]

(a) $K_3[Fe(CN)_6] > CaCl_2 > NaCl >$ glucose

(b) glucose > NaCl > $CaCl_2 > K_3[Fe(CN)_6]$

(c) NaCl > glucose > $K_3[Fe(CN)_6] > CaCl_2$

(d) $CaCl_2$ > NaCl > glucose > $K_3[Fe(CN)_6]$

(ii) Which one of the following is NOT a pyrimidine base? *(Recall)* [1]

(a) Uracil

(b) Thymine

(c) Cytosine

(d) Guanine

(iii) An excess of $AgNO_3$ solution is added to 100 mL of a 0·01 M solution of dichlorotetraaquachromium(III) chloride. The number of moles of AgCl precipitated would be: *(Apply)* [1]

(a) 0·003

(b) 0·01

(c) 0·001

(d) 0·002

(iv) The following data were obtained for the reaction $2X + Y \rightarrow$ products at a fixed temperature: [1]

Experiment	**[X] (mol L^{-1})**	**[Y] (mol L^{-1})**	**Initial rate (mol L^{-1} s^{-1})**
1	0·10	0·10	$1·0 \times 10^{-3}$
2	0·20	0·10	$4·0 \times 10^{-3}$
3	0·20	0·20	$8·0 \times 10^{-3}$

The rate law for this reaction is:

(Evaluate)

(a) rate = $k[X]^2[Y]$

(b) rate = $k[X][Y]^2$

(c) rate = $k[X]^2[Y]^2$

(d) rate = k[X][Y]

(v) Which one of the following amines will NOT give the carbylamine test with chloroform and alcoholic KOH? *(Understand)* [1]

(a) CH_3NH_2

(b) $C_6H_5NH_2$

(c) $(CH_3)_2NH$

(d) $C_2H_5NH_2$

(vi) Given below are two statements marked Assertion and Reason. Read the two statements carefully and select the correct option. [1]

Assertion: Aliphatic alcohols are weaker acids than water.

Reason: The alkoxide ion is stabilised by the electron-releasing (+I) effect of the alkyl group.

(Analyse)

(a) Both Assertion and Reason are true, and Reason is the correct explanation for Assertion.

(b) Both Assertion and Reason are true, but Reason is not the correct explanation for Assertion.

(c) Assertion is true and Reason is false.

(d) Both Assertion and Reason are false.

(vii) Given below are two statements marked Assertion and Reason. Read the two statements carefully and select the correct option. [1]

Assertion: The molar conductivity of acetic acid rises very sharply on dilution, whereas that of KCl rises only gradually.

Reason: The degree of dissociation of a weak electrolyte increases on dilution.

(Analyse)

(a) Both Assertion and Reason are true, and Reason is the correct explanation for Assertion.

(b) Both Assertion and Reason are true, but Reason is not the correct explanation for Assertion.

(c) Assertion is true and Reason is false.

(d) Both Assertion and Reason are false.

(C) Read the passage carefully and answer the questions that follow.

Alcohols and phenols are hydroxy derivatives of hydrocarbons. Ethanol is the most familiar alcohol: it is the active ingredient of hand sanitisers and is blended with petrol as a fuel. The boiling points of alcohols are much higher than those of ethers of comparable molar mass, because alcohol molecules are associated through hydrogen bonds. Aliphatic alcohols are very weak acids, weaker even than water. Phenol is a stronger acid than the alcohols, and the presence of an electron-withdrawing group on the ring increases its acidity further. The following data are taken from a chemistry data book.

Compound	**Ethanol**	**Methoxymethane**	**Propan-1-ol**	**Propan-2-ol**
Boiling point (K)	351	248	370	356

Compound	**Ethanol**	**Water**	**Phenol**	***p*-Cresol**	***m*-Nitrophenol**	***p*-Nitrophenol**
pK_a	15·9	15·7	10·0	10·3	8·4	7·2

Alcohols are classified as primary, secondary and tertiary, and the class can be recognised by the rate at which they react with the Lucas reagent.

(i) Ethanol and methoxymethane are isomers, yet ethanol boils at 351 K and methoxymethane at only 248 K. Explain. *(Understand)* [1]

(ii) Arrange ethanol, phenol and *p*-nitrophenol in the increasing order of acidic strength and give one reason why *p*-nitrophenol is more acidic than phenol. *(Analyse)* [1]

(iii) Name the reagent that can be used to distinguish propan-1-ol from propan-2-ol and state what is observed with each at room temperature. *(Apply)* [1]

SECTION B – 20 MARKS

Question 2

The standard reduction potentials of some electrodes are given below:

Electrode	**Fe^{2+}/Fe**	**Mg^{2+}/Mg**	**$I_2/2I^-$**	**Ag^+/Ag**
E° (V)	−0·44	−2·37	+0·54	+0·80

(i) Which of the species Fe^{2+}, Mg^{2+} and Ag^+ can be used to convert iodide ions to iodine? *(Analyse)* [1]

(ii) Which of the following reactions is spontaneous? Give a reason. [1]

I: $Fe^{2+} + 2Ag \rightarrow 2Ag^+ + Fe$

II: $Fe^{2+} + Mg \rightarrow Mg^{2+} + Fe$ *(Analyse)*

Question 3

An organic compound [A], $C_7H_6O_2$, on treatment with $SOCl_2$ gives compound [B]. On heating with NH_3, [B] gives compound [C]. [C] reacts with Br_2/KOH to give compound [D], which responds to the carbylamine test. Identify the compounds [A], [B], [C] and [D]. *(Understand)* [2]

Question 4

(i) Neha and Imran were given an assignment on the coordination complex ion $[Fe(CN)_6]^{4-}$ (atomic number of Fe = 26). Neha found that it has sp^3d^2 hybridisation and is paramagnetic, whereas Imran found that it has d^2sp^3 hybridisation and is diamagnetic. Evaluate the responses of Neha and Imran. *(Evaluate)* [2]

OR

(ii) Answer the following using crystal field theory for an octahedral complex.

(a) Write the electronic configuration of (1) a d^4 ion in a high-spin octahedral complex and (2) a d^6 ion in a low-spin octahedral complex, in terms of t_{2g} and e_g. *(Understand)* [1]

(b) Calculate the crystal field stabilisation energy (CFSE) of each of the two ions in terms of Δ_o. (Ignore the pairing energy.) *(Apply)* [1]

Question 5

Carbon dating of a bone fragment shows that it contains only 20 % of the carbon-14 that a living animal contains. Find the age of the bone. (Half-life of carbon-14 = 5730 years) *(Apply)* [2]

Question 6

How can the following conversions be carried out? (Write the chemical equations.)

(i) Nitrobenzene to benzenediazonium chloride *(Recall)* [1]

(ii) Chlorobenzene to biphenyl *(Recall)* [1]

Question 7

Substantiate the following statements with reasons.

(i) Cr^{2+} is a reducing agent but Mn^{3+} is an oxidising agent, although both have the d^4 configuration. (Atomic number of Cr = 24, Mn = 25) *(Analyse)* [1]

(ii) Zirconium (Z = 40) and hafnium (Z = 72) have very similar properties and are difficult to separate. *(Understand)* [1]

Question 8

Write the Nernst equation and calculate the emf of the following cell at 298 K: [2]

$$Al(s) \mid Al^{3+}(0{\cdot}01\ M) \parallel Cu^{2+}(0{\cdot}1\ M) \mid Cu(s)$$

Given: $E°(Al^{3+}/Al) = -1{\cdot}66$ V, $E°(Cu^{2+}/Cu) = +0{\cdot}34$ V *(Apply)*

Question 9

(i) Write one chemical equation to illustrate each of the following named reactions:

(a) Carbylamine reaction *(Recall)* [1]

(b) Sandmeyer reaction *(Recall)* [1]

OR

(ii) An organic compound [A], C_2H_7N, reacts with nitrous acid to give a compound [B] with the evolution of nitrogen. [B] on controlled oxidation gives [C], which reduces Tollens' reagent. Identify [A], [B] and [C] and write the equation for the reaction of [A] with nitrous acid. *(Understand)* [2]

Question 10

The rate of a reaction becomes four times when the temperature is raised from 293 K to 313 K. Calculate the activation energy of the reaction. What will be the value of the rate constant at very high temperature? ($R = 8{\cdot}314\ J\ K^{-1}\ mol^{-1}$) *(Apply)* [2]

Question 11

A 2 molal aqueous solution of sodium chloride shows an elevation in boiling point of 1·88 K. Calculate the van't Hoff factor. What does this value signify? (K_b of water = $0{\cdot}52\ K\ kg\ mol^{-1}$) *(Analyse)* [2]

SECTION C – 21 MARKS

Question 12

Write the chemical equations for the following reactions and name the products:

(i) Chlorobenzene is heated with conc. HNO_3 in the presence of conc. H_2SO_4. *(Apply)* [1]

(ii) Ethyl alcohol is treated with thionyl chloride in the presence of pyridine. *(Apply)* [1]

(iii) Chloroform is slowly oxidised by air in the presence of sunlight. *(Apply)* [1]

Question 13

The following data were obtained for the reaction A + B → products at 298 K:

Experiment	[A] (mol L^{-1})	[B] (mol L^{-1})	Rate (mol L^{-1} s^{-1})
1	0·10	0·10	$2{\cdot}0 \times 10^{-3}$
2	0·20	0·10	$4{\cdot}0 \times 10^{-3}$
3	0·10	0·20	$8{\cdot}0 \times 10^{-3}$

(i) What is the order of the reaction with respect to A and with respect to B? *(Analyse)* [1]

(ii) Calculate the overall order and the rate constant. *(Apply)* [1]

(iii) Calculate the rate of the reaction when [A] = $0{\cdot}50\ mol\ L^{-1}$ and [B] = $0{\cdot}30\ mol\ L^{-1}$. *(Apply)* [1]

Question 14

(i) Write the structural formula of the major organic product formed in each of the following:

(a) Ethanol is heated with excess conc. H_2SO_4 at 443 K. *(Apply)* [1]

(b) $CH_3OCH_2CH_3$ is heated with HI at 373 K. *(Apply)* [1]

(c) Phenol is treated with dilute nitric acid at low temperature. *(Apply)* [1]

OR

(ii) How will you convert the following? (Write chemical equations.)

(a) Ethyl bromide to diethyl ether *(Recall)* [1]

(b) Chlorobenzene to phenol *(Recall)* [1]

(c) Ethanal to propan-2-ol *(Recall)* [1]

Question 15

(i) Complete and balance: $KMnO_4 + H_2SO_4 + H_2C_2O_4 \rightarrow$ ____ + ____ + ____ + ____ *(Understand)* [1]

(ii) Complete and balance: $K_2Cr_2O_7 + H_2S + H_2SO_4 \rightarrow$ ____ + ____ + ____ + ____ *(Understand)* [1]

(iii) Zinc salts are white whereas cupric salts are coloured. Explain. *(Analyse)* [1]

Question 16

(i) Show the peptide linkage in glycylalanine, given the structures of glycine, $H_2N–CH_2–COOH$, and alanine, $H_2N–CH(CH_3)–COOH$. *(Understand)* [1]

(ii) Name the vitamins whose deficiency causes (a) beri-beri (b) scurvy. *(Recall)* [1]

(iii) Justify, with the help of a chemical reaction, that glucose is an aldose. *(Understand)* [1]

Question 17

(i) Three electrolytic cells containing solutions of $AgNO_3$, $CuSO_4$ and $ZnSO_4$ are connected in series. A steady current is passed until 1·62 g of silver has been deposited at the cathode of the first cell. (Atomic weights: Ag = 108, Cu = 63·5, Zn = 65·3; 1 F = 96500 C)

(a) How much charge was passed through the electrolyte? *(Apply)* [1]

(b) What mass of copper and of zinc is deposited at the respective cathodes? *(Apply)* [2]

OR

(ii) The specific conductance of a 0·020 M solution of acetic acid at 298 K is $2·34 \times 10^{-4}$ S cm^{-1}. The molar conductivities at infinite dilution of H^+ and CH_3COO^- ions are 349·1 and 40·9 S cm^2 mol^{-1} respectively. Calculate:

(a) the molar conductance of the solution *(Apply)* [1]

(b) the degree of dissociation of acetic acid *(Apply)* [1]

(c) the dissociation constant of acetic acid *(Apply)* [1]

Question 18

Identify the compounds [A], [B] and [C] in each of the following sequences:

(i) $C_6H_5NO_2 \xrightarrow{Fe/HCl} [A] \xrightarrow[273–278\ K]{NaNO_2 + HCl} [B] \xrightarrow{KI} [C]$ [1½]

(Understand)

(ii) $C_6H_5N_2^+Cl^- \xrightarrow{CuCN,\ \Delta} [A] \xrightarrow{H_2O/H^+} [B] \xrightarrow{NH_3,\ \Delta} [C]$ [1½]

(Understand)

SECTION D – 15 MARKS

Question 19

(i) What happens when (write the chemical equations):

(a) Calcium acetate is subjected to dry distillation. *(Recall)* [1]

(b) Propanone is treated with zinc amalgam and concentrated hydrochloric acid. *(Recall)* [1]

(c) Acetic acid is heated with ethyl alcohol in the presence of conc. H_2SO_4. *(Recall)* [1]

(ii) An organic compound [X], C_2H_4O, gives a yellow precipitate when heated with iodine and sodium hydroxide. [X] on oxidation forms [Y] and on treatment with HCN forms [Z], which on hydrolysis gives 2-hydroxypropanoic acid.

(a) Write the structures of [X], [Y] and [Z]. *(Understand)* [1]

(b) Name the product formed when [X] is treated with dilute NaOH and write the equation. [1]
(Apply)

OR

(iii) What happens when (write the chemical equations):

(a) Benzaldehyde is treated with HCN and the product is hydrolysed. *(Recall)* [1]

(b) Formaldehyde is heated with concentrated NaOH. *(Recall)* [1]

(c) Acetyl chloride is hydrogenated over Pd supported on $BaSO_4$. *(Recall)* [1]

(iv) A ketone [A], C_4H_8O, undergoes the haloform reaction and on reduction gives [B]. [B] on heating with conc. H_2SO_4 at 443 K gives [C], which on ozonolysis with Zn/H_2O gives [D] only. Identify [A], [B], [C] and [D]. *(Analyse)* [2]

Question 20

(i) When $AgNO_3$ solution is added to a solution of the coordination compound $PdCl_2 \cdot 4NH_3$, two moles of AgCl are precipitated per mole of the compound. On this basis designate: (a) the coordination entity (b) the counter ions (c) the coordination number of the central metal ion (d) the oxidation number of Pd. *(Apply)* [2]

(ii) Write the IUPAC name of (a) $[PtCl_2(NH_3)_4][PtCl_4]$ and (b) $K_2[Zn(OH)_4]$. *(Recall)* [1½]

(iii) The complex $CoClSO_4 \cdot 5NH_3$ exists in two isomeric forms [A] and [B]. [A] gives a white precipitate with $AgNO_3$ but not with $BaCl_2$; [B] gives a white precipitate with $BaCl_2$ but not with $AgNO_3$. Identify [A] and [B] and name the type of isomerism. *(Analyse)* [1½]

Question 21

(i) Answer the following:

(a) A solution prepared by dissolving 5·60 g of a non-volatile solute in 25·0 g of chloroform boils at 68·04 °C. Calculate the molar mass of the solute. (The boiling point of pure chloroform is 61·7 °C and K_b for chloroform is 3·63 K kg mol^{-1}.) *(Apply)* [2]

(b) A 0·15 M aqueous solution of KCl exerts an osmotic pressure of 6·8 atm at 310 K. Calculate the van't Hoff factor. (R = 0·0821 L atm K^{-1} mol^{-1}) *(Apply)* [1]

(c) A student gave the following decreasing order of depression in freezing point for 0·1 molal solutions: urea > NaCl > $MgCl_2$. Is the order correct? Give a reason. *(Evaluate)* [1]

(d) Which colligative property is preferred for the determination of the molar mass of macromolecules such as proteins? Give a reason. *(Understand)* [1]

OR

(ii) Answer the following:

(a) Addition of NaCl or glucose to water raises its boiling point, but addition of methanol lowers it. Give a reason. *(Analyse)* [1]

(b) Why does a soft drink bottle give a fizzing sound when it is opened? *(Understand)* [1]

(c) The vapour pressures of pure liquids A and B are 450 and 700 mm Hg respectively at 350 K. Find the composition of the liquid mixture if the total vapour pressure is 600 mm Hg, and the composition of the vapour phase. *(Apply)* [3]

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